Project Report On

MeshComm: Emergency Communication System

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

Aaromal M (U2203001)
Aaron Korah Mathew (U2203005)
Alan Madhu Kuriakose (U2203022)
Amal Zachariah Mathew (U2203035)

Under the guidance of
Mr. Tonny Binoy

Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039
March 2026

CERTIFICATE

This is to certify that the project report entitled ***MeshComm: Emergency Communication System*** *is a bonafide record of the work done by* ***Aaromal M (U2203001), Aaron Korah Mathew (U2203005), Alan Madhu Kuriakose (U2203022), Amal Zachariah Mathew (U2203035)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

Project Guide
Mr.Tonny Binoy
Assistant Professor
Dept. of CSE
RSET

Project Co-ordinator
Dr. Jisha G
Associate Professor
Dept. of CSE
RSET

Dr. Preetha K.G
Professor
Dept. of CSE
RSET

ACKNOWLEDGMENT

I wish to express my sincere gratitude towards **Rev.Dr.Jaison Paul Mulerikkal CMI**, Principal of RSET, and **Dr.Preetha K.G**, Head of the Department of Computer Science and Engineering for providing me with the opportunity to undertake my project, MeshComm: Emergency Communication System.

I am highly indebted to my project coordinator, **Dr.Jisha G**, Associate Professor, Computer Science and Engineering, for her valuable support.

It is indeed my pleasure and a moment of satisfaction for me to express my sincere gratitude to my project guide **Mr.Tonny Binoy** for his patience and all the priceless advice and wisdom he has shared with me.

Last but not the least, I would like to express my sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Aaromal M

Aaron Korah Mathew

Alan Madhu Kuriakose

Amal Zachariah Mathew

Abstract

In disaster situations, cellular networks and Internet connectivity often fail, leaving individuals and communities without a means to communicate or request help. This work presents a lightweight, smartphone-based communication system that enables offline message exchange using Bluetooth Low Energy (BLE) technology without completely relying on the traditional infrastructure.

The system employs a simple flooding mechanism in which each device rebroadcasts the received packets with a bounded time-to-live (TTL), enabling multi-hop propagation across nearby devices. This work also incorporates Delay Tolerant Network (DTN) principles for improving the chances of encountering potential gateway nodes. Two types of broadcasts are supported: (1) user-generated text messages for short-range communication with nearby users and (2) GPS coordinate packets that propagate through the mesh until they reach a device with Internet access. Once a gateway device is encountered, it automatically forwards the GPS message frame to a cloud server which can be accessed by emergency services, thus enabling the emergency services or responders to locate the affected users even when direct connectivity is unavailable or not reliable. This implementation uses Android's standard BLE advertising and scanning capabilities. It also uses a compact frame-encoding scheme and a foreground service to maintain continuous operation even when the devices are mobile.

One constant for the successful execution of this work is the possession of a mobile device which is enabled with the app. The app must be in an active state for continuous broadcasting. Experimental evaluation confirmed that the prototype enabled message relaying across multiple devices without Wi-Fi or mobile data, thereby demonstrating its potential as a resilient communication method for emergency response and in low-resource environments. This work highlights the feasibility of using BLE-based simple flooding on

common smartphones to create a practical, energy-efficient offline mesh communication network.

Contents

Acknowledgment **i**

Abstract **ii**

List of Abbreviations **vii**

List of Figures **viii**

List of Tables **ix**

1 Introduction **1**

1.1 Background 1

1.2 Problem Definition 3

1.3 Scope and Motivation 3

1.4 Objectives 4

1.5 Challenges 4

1.6 Assumptions 5

1.7 Societal / Industrial Relevance 5

1.8 Organization of the Report 5

1.9 Summary of Chapter 6

2 Literature Survey **7**

2.1 Review of Key Research Works 7

2.1.1 Understanding the Performance of Bluetooth Mesh: Reliability, Delay, and Scalability Analysis -Rondón et al., (2019) 7

2.1.2 A Comprehensive Evaluation of Bluetooth Low Energy Mesh - Zhao et al., (2024) 8

2.1.3 A Rapid Flooding Approach Based on Adaptive Delay and Low-Power Sleep for BLE Mesh Networks - Wang et al., (2024) 9

2.1.4 Combining Software-Defined and Delay-Tolerant Networking Concepts With Deep Reinforcement Learning to Enhance Vehicular Networks - Nakayima et al., (2024) 10
2.1.5 Collective Victim Counting in Post-Disaster Response: A Distributed, Power-Efficient Algorithm via BLE Spontaneous Networks - Longo et al., (2025) 10
2.2 Summary and Gaps Identified 12
2.2.1 Summary 12
2.2.2 Gaps Identified 13
2.3 Summary of Chapter 14

3 System Design **15**
3.1 Overall System Architecture 15
3.2 Component Design 16
3.2.1 User Interface Component 18
3.2.2 Messsage Processing Component 18
3.2.3 BLE Communication Component 18
3.2.4 Gateway Component 19
3.2.5 Cloud Server Component 19
3.3 Module Division 19
3.3.1 User Interface Module 19
3.3.2 Message Processing Module 20
3.3.3 BLE Communication Module 20
3.3.4 Gateway Module 20
3.3.5 Cloud Server Module 21
3.4 Tools and Technologies 22
3.4.1 Software Requirements 22
3.4.2 Hardware Requirements 23
3.5 Work breakdown and Responsibilities 24
3.5.1 Module 1: User Interface Module - Alan, Aaron, Aaromal 24
3.5.2 Module 2: BLE Communication Module - Alan, Aaron 24
3.5.3 Module 3: Message Processing Module - Alan, Aaron 24

3.5.4 Module 4: Gateway Module - Amal, Aaromal 24
3.5.5 Module 5: Cloud Server Module - Amal, Aaromal 25
3.6 Key Deliverables 25
3.7 Project Timeline 26
3.8 Summary of chapter 26

4 Results and Discussions **27**
4.1 System Implementation and Testing Setup 27
4.2 Message Transmission and Multi-Hop Communication Results 28
4.3 GPS Alert Broadcasting and Gateway Functionality 28
4.4 Observation and Performance Analysis 29
4.5 Conclusion 30

5 Conclusions & Future Scope **31**
5.1 Conclusion 31
5.2 Future Scope 31

References **33**

List of Publications **35**

Appendix A: Presentation **36**

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes **55**

Appendix C: CO-PO-PSO Mapping **60**

List of Abbreviations

Acronym	Expansion
BLE	Bluetooth Low Energy
DTN	Delay-Tolerant Networking
TTL	Time To Live
IoT	Internet of Things
PDU	Protocol Data Unit
QoS	Quality of Service
RSSI	Received Signal Strength Indicator
ADV	Advertisement
GATT	Generic Attribute Profile
API	Application Programming Interface
MQTT	Message Queueing Telemetry Transport
SDN	Software Defined Networking
VANET	Vehicular Ad-hoc Network
DB	Database
UID	Unique Identifier
CPU	Central Processing Unit
SDK	Software Development Kit
GPS	Global Positioning System
UI	User Interface
HTTPS	Hypertext Transfer Protocol Secure

List of Figures

3.1 System Architecture 16
3.2 System Flowchart 17
3.3 System Flow Diagram 21
3.4 Gantt Chart 26

List of Tables

2.1 Comparison table 12

Chapter 1

Introduction

1.1 Background

Communication failure is one of the most critical problems faced in disaster-affected regions. During disasters like earthquakes, cyclones, landslides or floods traditional communication infrastructure like cellular towers and the internet backbone often gets damaged or overloaded making them very unreliable. For instance, in the 2015 Nepal Earthquake, widespread tower failures left millions of users unable to send distress messages, and the 2022 Kerala floods caused recurrent network outages that hindered rescue operations. One other instance was when victims of 2024 Wayanad landslides experienced delayed response from the emergency responders due to total wipeout of the communication infrastructure in the area. The inability of the survivors and emergency responders to communicate effectively can delay life-saving interventions and increase casualties.

However, even when mobile networks fail, most survivors would still possess a functioning smartphone. These devices provide essential capabilities independent of the internet, such as the Bluetooth Low Energy (BLE) for short-range wireless communication and built-in GPS sensors for location tracking. This creates a unique opportunity to enable emergency communication using only the devices already available to people in the disaster zones.

There exist several technologies and applications that demonstrate the feasibility of infrastructure independent wireless communication systems over short ranges. These systems provide a useful context for the proposed work despite their design goals and operational assumptions differing significantly from emergency-oriented BLE communication system.

Smartphone-based offline messaging applications such as FireChat and Bridgefy utilize Bluetooth and, in some cases Wi-Fi Direct to enable peer-to-peer communication without internet connectivity. While they demonstrate the practicality of device-to-device messaging, they primarily focus on short-range chat functionality and do not provide bounded flooding control, message lifetime management, or mechanisms for escalating emergency information beyond the local mesh. They also do not have any provisions for handling delayed exposure to networks (DTN support).

Applications like Briar which are privacy-focused extend offline communication by supporting Bluetooth and Wi-Fi-based message synchronization with strong cryptographic guarantees. However, such systems prioritize security and anonymity over rapid dissemination, often introducing higher computational and energy overhead. Additionally, they are not designed to support opportunistic forwarding of GPS-based emergency alerts to external infrastructure.

Disaster-oriented communication platforms such as Serval Mesh employ mobile ad hoc networking to support text and voice communication in disconnected environments. Although these systems do align more closely with emergency use cases, they usually rely on connection-oriented communication and more complex network configuration, thus limiting their suitability for lightweight BLE-based deployment on commodity smartphones.

In contrast to the aforementioned applications, the system proposed in this work deliberately adopts a minimalist design philosophy. It uses BLE advertising for connectionless communication, bounded folding with Time-To-Live (TTL), delay tolerance and opportunistic gateway-based cloud forwarding. The proposed approach avoids routing tables and persistent connections. The design prioritizes rapid deployability, robustness under mobility and suitability for emergency scenarios where simplicity and resilience are critical.

This project presents a BLE-based emergency communication system that enables smartphones to exchange rescue messages and GPS coordinates even without cellular or internet connectivity. The system uses a simple flooding mechanism, where each device rebroadcasts a newly received message once and is controlled by a Time-To-Live

(TTL) value. This approach ensures that emergency messages propagate quickly across neighbouring devices without requiring routing tables, network topology awareness, or complex mesh protocols.

1.2 Problem Definition

Design an energy efficient and delay-tolerant emergency communication system using Bluetooth Low Energy with Opportunistic Gateway Finder that guarantees eventual message delivery even under sparse connectivity and network partitioning.

1.3 Scope and Motivation

The project focuses on developing and evaluating an Emergency Communication System using Bluetooth Low Energy (BLE) with an Opportunistic Gateway Finder that enables reliable message exchange during disasters or network outages without a complete dependence on internet or cellular infrastructure. It integrates multi-hop BLE mesh networking, store-carry-forward (DTN) routing, and local buffer repositories on each node to ensure delay-tolerant and energy-efficient communication even under sparse connectivity.It caters to the need for a resilient, low-cost, and infrastructure-independent communication solution, the system allows survivors to transmit GPS location, SOS messages, and updates during emergencies. Featuring an Android-based interface for real-time messaging and an opportunistic Internet Gateway for cloud synchronization when connectivity is restored, the project aims to enhance disaster response, ensure message reliability, and maintain seamless communication continuity in challenging environments.

During natural disasters, network infrastructures often collapse, leaving affected populations without any means of communication. Emergency messages become critical for survival, yet existing communication systems rely heavily on centralized networks. Leveraging BLE and DTN principles provides a resilient, decentralized method for message delivery even in the absence of internet or cellular connectivity.

1.4 Objectives

- Main Goal: Build a resilient emergency communication system using BLE + DTN principles that runs on commodity smartphones.
- Specific Goals:
- Multi-hop Communication: Messages should propagate device-to-device until reaching recipient.
- Delay Tolerance: Packets are stored in local buffer and forwarded when peers appear.
- Proximity Communication: Peer to peer communication using text messages for devices in range.
- GPS location broadcast: Broadcast GPS location as a distress signal supported by an Opportunistic Gateway Finder
- Opportunistic Internet Gateway: Auto-upload messages to a cloud server once any device regains Internet connectivity.
- Deduplication: Prevent redundant rebroadcasting of same message
- Energy Optimization: Minimize BLE advertising/scan intervals while ensuring reliability.
- Responder Dashboard: Provide Emergency Services a dashboard that helps them locate distress signals received by the cloud server

1.5 Challenges

- Limited BLE range and frequent packet collisions leading to unreliable data transfer.
- High energy consumption due to continuous scanning and advertising operations.
- Difficulty in ensuring users keep the application active during emergencies.
- Possibility of false or malicious alerts disrupting network integrity.
- Balancing reliability and energy efficiency while designing scalable DTN buffers for large networks.

1.6 Assumptions

- All participating mobile devices are equipped with Bluetooth Low Energy (BLE 5.0 or higher) hardware.
- Users (survivors, responders, or volunteers) have installed and enabled the mobile application and the device stays powered on with Bluetooth and occasionally GPS kept active.
- At least one device occasionally gains Internet access to upload buffered messages to the cloud.
- Devices have sufficient battery power to perform periodic BLE advertising and scanning.
- Local device storage (buffer) is sufficient to hold undelivered messages temporarily.

1.7 Societal / Industrial Relevance

Disasters often lead to total communication breakdowns, isolating victims and rescue teams. In such situations, the proposed system provides a lifeline for information exchange without relying on Internet or cellular networks. By enabling short-range, peer-to-peer communication through Bluetooth Mesh and Delay-Tolerant Networking (DTN), this project directly contributes to public safety, disaster resilience, and emergency response coordination. The system aligns with the growing interest in infrastructure-independent IoT communication and resilient network design. Industries focusing on telecommunications, IoT, defense, and public safety can benefit from the underlying architecture.

1.8 Organization of the Report

The report is organized as follows:

- **Chapter 1: Introduction** – Overview of MeshComm, project motivation, scope, challenges, assumptions, and societal relevance.
- **Chapter 2: Literature Survey** – Analysis of related Bluetooth Mesh Communication systems and industry best practices.

- **Chapter 3: System Design** – Description of MeshComm's modules, architecture, tools and technologies used.
- **Chapter 4: Results and Discussion** – Summary of findings, effectiveness of MeshComm and its limitations
- **Chapter 5: Conclusion and Future Scope**

1.9 Summary of Chapter

This chapter provided a comprehensive overview of MeshComm: Emergency Communication System. It included the background and motivation for the project. The section that followed detailed the scope and objectives. The chapter also describes the challenges, assumptions and societal relevance associated with the project.

Chapter 2

Literature Survey

The effectiveness of any communication system, particularly in critical environments such as disaster response, depends heavily on its underlying network architecture, routing efficiency, scalability, and reliability. To design an Emergency Communication System using Bluetooth Mesh and Delay-Tolerant Networking (DTN), it is essential to understand existing research on Bluetooth Mesh performance optimization, adaptive flooding mechanisms, DTN routing techniques, and real-world deployment strategies.

This chapter reviews several key journal articles and research studies that provide the theoretical foundation and technological insight for our project. The selected works cover diverse but interrelated domains including BLE Mesh reliability and delay analysis, adaptive and power-efficient flooding algorithms, integration of DTN with intelligent control mechanisms, and distributed BLE-based communication models for disaster management.

2.1 Review of Key Research Works

2.1.1 Understanding the Performance of Bluetooth Mesh: Reliability, Delay, and Scalability Analysis -Rondón et al., (2019)

- **Core Contribution:**This paper provides an extensive evaluation of Bluetooth Mesh performance, focusing on the interdependence of protocol parameters that influence end-to-end reliability, delay, and scalability. It presents one of the first studies to analyze how advertising and scanning configurations affect Quality-of-Service (QoS) and offers clear guidelines for optimizing Bluetooth Mesh in large-scale IoT networks.

- **Methodology:** The authors conducted extensive simulations and real-world experiments to evaluate how advertising and scanning intervals, randomization, and interference affect network performance. Parameters such as ScanInterval and Tin-

terPDU were systematically varied under different conditions, including the presence of WLAN interference, to observe their impact on latency and reliability across multiple hops.

- **Key Findings:** Results demonstrated a strong dependency between advertising and scanning configurations. Introducing randomized intervals minimized flooding collisions and reduced latency to around 250 ms for 10-hop transmissions. However, scalability challenges were identified due to broadcast storms in dense topologies. The study also confirmed Bluetooth Mesh's susceptibility to external interference, even under restricted channel operation.

- **Relevance to Project:** This paper provides valuable insights into optimizing BLE Mesh for delay-tolerant and interference-prone environments. Its findings on protocol parameter tuning and scalability directly support the design of our Emergency Communication System, helping ensure efficient message propagation and reliable multi-hop communication during critical operations.

2.1.2 A Comprehensive Evaluation of Bluetooth Low Energy Mesh - Zhao et al., (2024)

- **Core Contribution:** This paper delivers an experimental performance evaluation of BLE Mesh using commercial off-the-shelf (COTS) devices. It investigates how network parameters such as transmission power, packet interval, and mesh topology affect performance, offering real-world evidence on BLE-Mesh behavior in IoT contexts.

- **Methodology:** The authors implemented BLE Mesh testbeds in indoor and outdoor environments and measured parameters like packet delivery ratio, transmission range, and network stability under various configurations. Experiments evaluated both single-hop and multi-hop scenarios to capture the relationship between node arrangement and communication efficiency.

- **Key Findings:** BLE Mesh achieved reliable multi-hop communication under optimized transmission power and packet intervals, but performance degraded significantly in irregular node placements. The study provided practical guidelines for

achieving optimal range, throughput, and stability in COTS-based BLE deployments.

- **Relevance to Project:** This work reinforces the practical viability of BLE Mesh for real-world deployments like emergency communication systems. The empirical observations guide the tuning of power and timing configurations in our proposed BLE-based framework to maximize coverage and maintain reliability under constrained environments.

2.1.3 A Rapid Flooding Approach Based on Adaptive Delay and Low-Power Sleep for BLE Mesh Networks - Wang et al., (2024)

- **Core Contribution:** The paper introduces the FADLS (Fast Adaptive Delay and Low-Power Sleep) algorithm to enhance BLE Mesh flooding efficiency. It aims to mitigate redundant transmissions and collisions while preserving fast propagation, thereby improving scalability and energy efficiency in dense networks.

- **Methodology:** The algorithm dynamically adjusts random relay delays and activates low-power relay sleep based on real-time neighbor information. Simulations were performed to evaluate its performance against existing flooding and probabilistic broadcast techniques under various network densities.

- **Key Findings:** FADLS achieved significant reductions in packet collisions and end-to-end delay compared to static flooding approaches. Its adaptive mechanism maintained rapid message propagation while conserving energy through selective relay activation. The lightweight neighbor discovery mechanism proved efficient for limited-payload BLE Mesh packets.

- **Relevance to Project:** This approach directly aligns with our objective of achieving energy-efficient and low-latency communication in emergency BLE Mesh networks. The adaptive flooding concept can enhance our project's store-and-forward performance while minimizing collisions across dynamically changing topologies.

2.1.4 Combining Software-Defined and Delay-Tolerant Networking Concepts With Deep Reinforcement Learning to Enhance Vehicular Networks - Nakayima et al., (2024)

- **Core Contribution:** This paper proposes an integration of Software-Defined Networking (SDN) and Delay-Tolerant Networking (DTN) principles through a Centralized Controller Multi-Agent (CCMA) Reinforcement Learning algorithm to enhance data delivery in Vehicular Ad-hoc Networks (VANETs).

- **Methodology:** The proposed CCMA algorithm dynamically manages routing protocols (e.g., Spray and Wait, Epidemic, PRoPHETv2) and buffer scheduling policies (FIFO, EDF, Random) based on network state parameters such as buffer utilization, TTL, and link quality. Simulation environments modeled three real-world scenarios: advanced, limited-resource, and opportunistic regions.

- **Key Findings:** The integrated SDN-DTN-RL system improved message delivery ratio, reduced latency, and optimized link quality across all environments compared to traditional single-protocol VANET systems. The model demonstrated adaptability to varying conditions, achieving higher reliability in disrupted or resource-constrained networks.

- **Relevance to Project:** The paper's hybrid control mechanism supports our research direction in designing resilient, adaptive BLE Mesh-based systems. The use of RL for dynamic resource management provides a reference framework for optimizing routing and buffer allocation within our emergency communication prototype.

2.1.5 Collective Victim Counting in Post-Disaster Response: A Distributed, Power-Efficient Algorithm via BLE Spontaneous Networks - Longo et al., (2025)

- **Core Contribution:** This paper presents a distributed algorithm for estimating the number of victims in disaster zones using BLE-based spontaneous networks. It enables infrastructure-free, privacy-preserving data exchange among mobile and wearable devices to assist first responders in large-scale emergencies.

- **Methodology:** The proposed method leverages Bluetooth Low Energy advertisements and Bloom filters for probabilistic counting of nearby devices. Experiments

were conducted in simulated post-disaster environments and validated through real-world measurements. The system's convergence speed and accuracy were evaluated across varying densities of participating devices.

- **Key Findings:**The distributed counting algorithm achieved high accuracy and rapid convergence even beyond direct BLE range through device-to-device relay. The use of random identifiers preserved user privacy while maintaining scalability and energy efficiency. The approach demonstrated strong potential for real-time victim estimation in disrupted communication environments.

- **Relevance to Project:**This paper's use of BLE spontaneous networks is directly relevant to our emergency communication framework. Its design principles for decentralized data exchange and privacy-preserving coordination support the long-term scalability, resilience, and practicality of our BLE Mesh-based disaster response system.

2.2 Summary and Gaps Identified

2.2.1 Summary

Table 2.1: Comparison table

Paper/Study	Approach	Domain	Strength	Limitation
Rondón et al. (2019)	Experimental analysis of BLE Mesh parameters	Wireless IoT Communication	Provides detailed insight into BLE Mesh reliability, latency, and scalability under interference and parameter variation	Limited scalability in dense networks; high susceptibility to 2.4 GHz interference
Zhao et al. (2024)	Empirical BLE Mesh evaluation using COTS devices	IoT System Performance	Practical results validating real-world BLE Mesh deployment, improved configuration guidelines for stability	Performance drops under irregular topologies and uncontrolled environments
Wang et al. (2024)	FADLS – Adaptive Delay and Low-Power Sleep Flooding Algorithm	Energy-efficient BLE Mesh Networking	Significant reduction in packet collisions and delay, high scalability and energy optimization	Complexity in adaptive parameter tuning; limited large-scale field validation

Nakayima et al. (2024)	SDN + DTN integration with Deep Reinforcement Learning	Vehicular Ad-hoc Networks	Enhanced data delivery ratio, reduced latency, robust multi-protocol routing under disruptions	Increased training complexity and overhead; depends on centralized coordination
Longo et al. (2025)	BLE-based distributed probabilistic counting via Bloom filters	Emergency Response and Crowd Estimation	Infrastructure-free, privacy-preserving and scalable victim counting; fast convergence and accuracy	Limited by BLE range in ultra-sparse scenarios; requires higher adoption density

2.2.2 Gaps Identified

The reviewed literature provides valuable insights into Bluetooth Low Energy (BLE) mesh networking, including performance evaluation, routing optimization, and energy-efficient communication mechanisms. However, several gaps remain. Many studies focus primarily on performance analysis and protocol optimization of BLE mesh networks rather than practical emergency communication applications. While algorithms such as adaptive flooding and congestion-aware routing improve network efficiency, they often introduce complex configuration or computational overhead, making them less suitable for lightweight deployment on standard smartphones.

Additionally, several approaches rely on centralized coordination or advanced machine learning techniques, which may not be feasible in disaster scenarios where infrastructure and computational resources are limited. Existing research also rarely integrates opportunistic gateway-based Internet forwarding, which is crucial for enabling distress messages to eventually reach rescue authorities when connectivity becomes available. Furthermore, most studies evaluate BLE mesh performance in controlled environments, leaving a gap in simple, deployable systems designed specifically for disaster communication using com-

modity devices.

Therefore, there is a need for a lightweight, infrastructure-independent communication system that uses simple flooding mechanisms, operates on widely available smartphones, and supports opportunistic forwarding of emergency location information to cloud-based services.

2.3 Summary of Chapter

This chapter presented a detailed review of research works related to Bluetooth Mesh networks and Delay-Tolerant Networking (DTN), both of which form the foundation of the emergency communication system. The reviewed studies collectively highlight key challenges and solutions in achieving reliable, scalable, and energy-efficient communication under conditions of limited connectivity.

Research on Bluetooth Mesh (Rondón et al., 2019; Zhao et al., 2024) emphasized the influence of advertising, scanning, and transmission parameters on network reliability and latency, providing valuable insights for optimizing protocol configurations in our design. Studies such as Wang et al. (2024) introduced adaptive flooding algorithms that significantly improve scalability and power efficiency — techniques that directly inform our system's message forwarding logic.

Further, works like Nakayima et al. (2024) demonstrated how Delay-Tolerant Networking and intelligent control mechanisms can enhance delivery ratio and adaptability in highly dynamic networks. Longo et al. (2025) showcased the use of BLE-based spontaneous networks for real-world disaster scenarios.

From these findings, the chapter establishes that combining Bluetooth Mesh and DTN principles can yield a resilient, decentralized communication framework capable of functioning in disrupted or infrastructure-less environments. These insights guide the architectural design, routing strategy, and optimization choices implemented in the subsequent chapters of this project.

Chapter 3

System Design

3.1 Overall System Architecture

The proposed system follows a hybrid - decentralized mesh communication architecture designed to operate without reliance on conventional network infrastructure. The system consists of multiple smartphone nodes that communicate with each other using Bluetooth Low Energy (BLE). Each device acts as both a transmitter and relay node, enabling messages to propagate through the network using a simple flooding mechanism controlled by a Time-To-Live (TTL) parameter. The architecture is composed of several functional modules including the User Interface Module, Message Processing Module, BLE Communication Module, Gateway Module, and Cloud Server Module.

The User Interface Module allows users to initiate communication by sending text messages or broadcasting GPS coordinates. The Message Processing Module encodes outgoing data into compact packets and decodes received packets while handling message identification and TTL management. The BLE Communication Module manages advertising and scanning operations to enable device discovery and packet exchange between nearby nodes. When a node with Internet connectivity is encountered, the Gateway Module forwards stored GPS distress messages to a remote Cloud Server. The Cloud Server Module stores and manages these alerts so that emergency responders can access the information for rescue coordination. This architecture enables resilient, infrastructure-independent communication while still allowing integration with centralized services when connectivity becomes available.

This integrated design ensures resilient, energy-efficient, and delay-tolerant communication, maintaining connectivity and message delivery even in disaster zones with limited

or no infrastructure.

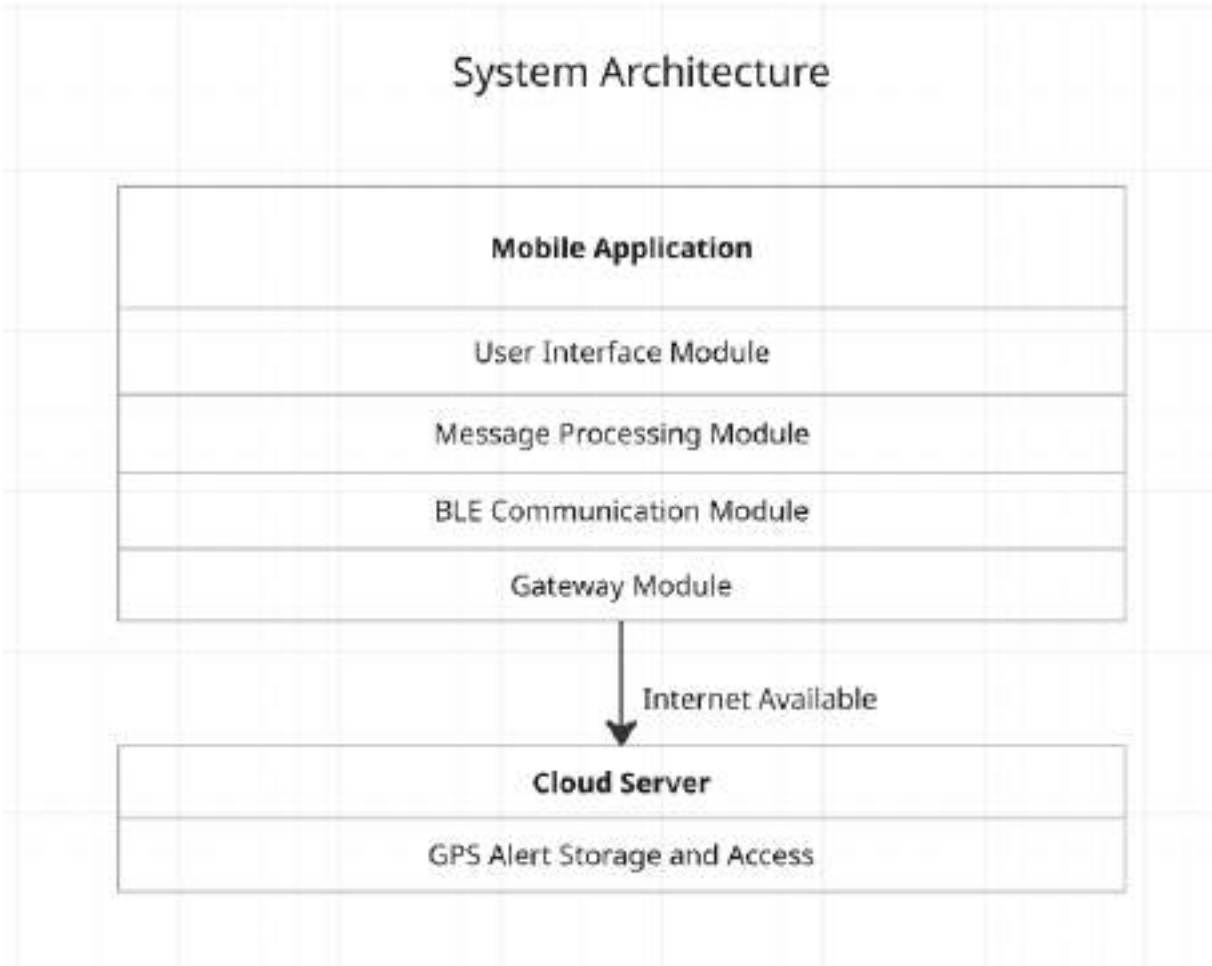


Figure 3.1: System Architecture

3.2 Component Design

The component design of the system focuses on defining the internal structure and functionality of each module involved in the mesh communication framework. Each component is designed to operate independently while interacting with other components to ensure reliable communication in infrastructure-less environments. The modular architecture improves system maintainability, scalability, and ease of development by separating user interaction, data processing, communication handling, and external connectivity into distinct functional units.

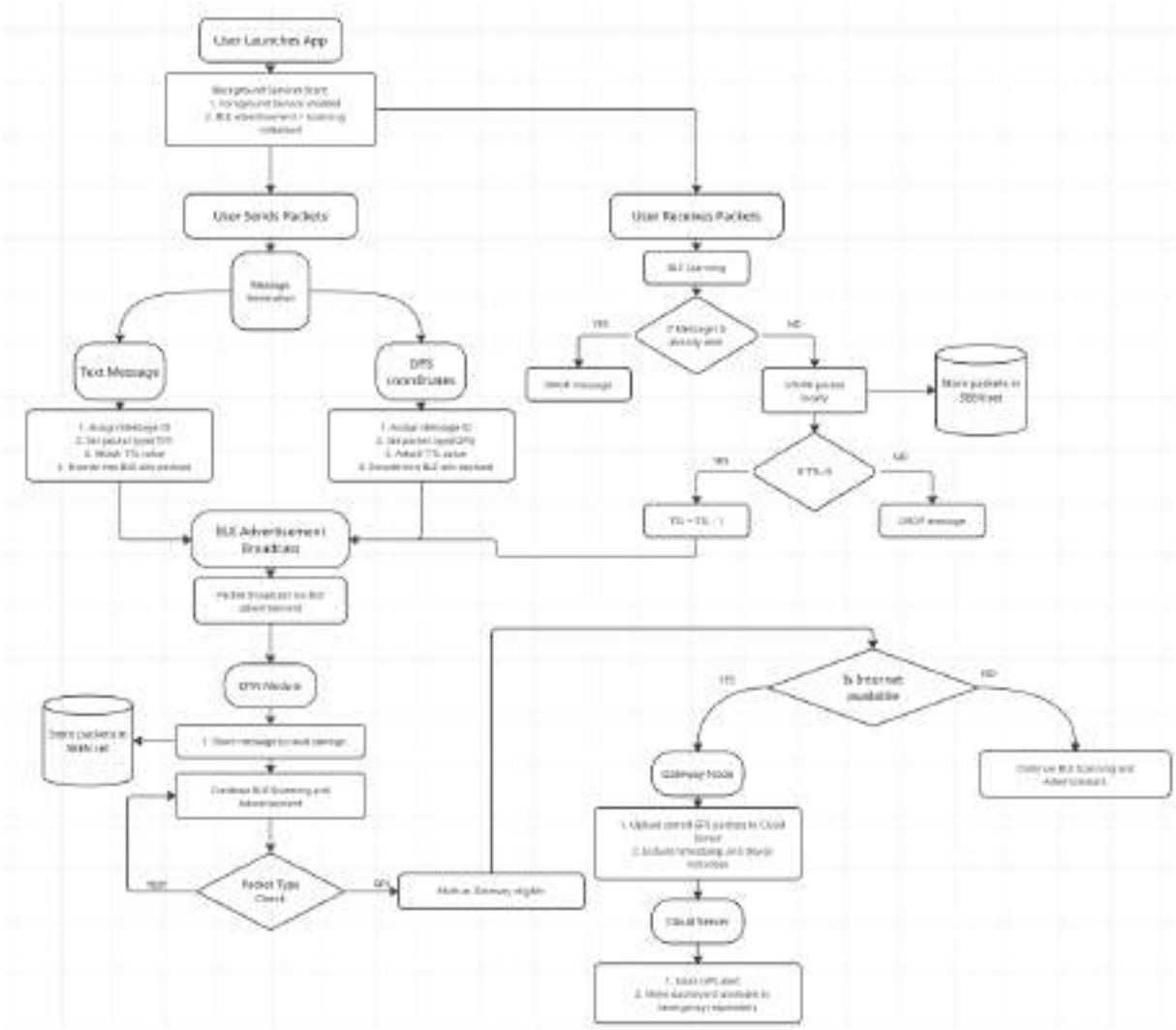


Figure 3.2: System Flowchart

3.2.1 User Interface Component

The User Interface component is responsible for handling user interactions and displaying system status information. It provides controls that allow users to start or stop the mesh communication service, send text messages, and broadcast their current GPS location. The interface also displays indicators showing the status of the mesh network and Internet connectivity, along with a scrollable activity log that records both sent and received messages. This component is implemented using Jetpack Compose, which supports reactive UI updates based on application state changes.

3.2.2 Messsage Processing Component

The Message Processing component manages the encoding, decoding, and validation of data packets transmitted within the mesh network. When a user sends a message or GPS alert, the component converts the data into a compact packet format that includes a message type identifier and a Time-To-Live (TTL) field. The TTL value ensures that messages propagate only a limited number of hops to prevent excessive network flooding. Upon receiving packets from nearby devices, this component decodes the packet, extracts relevant data, and determines whether the message should be displayed or forwarded further in the network.

3.2.3 BLE Communication Component

The BLE Communication component handles all Bluetooth Low Energy operations required for mesh communication. It continuously advertises outgoing packets while simultaneously scanning for incoming packets from neighboring devices. When a packet is detected, the component forwards it to the Message Processing module for interpretation. If the packet is valid and has remaining TTL, the communication component rebroadcasts it to extend its propagation range. This component operates through a foreground service to ensure uninterrupted communication even when the application is running in the background.

3.2.4 Gateway Component

The Gateway component is responsible for connecting the offline mesh network with external Internet services when connectivity becomes available. It continuously monitors the device's network status to detect whether Wi-Fi or mobile data is active. When Internet connectivity is detected, the gateway retrieves stored GPS alerts from the mesh network and forwards them to a remote cloud server. This component follows a store-carry-forward mechanism inspired by delay-tolerant networking, ensuring that emergency alerts can eventually reach rescue authorities even if the originating device does not have direct Internet access.

3.2.5 Cloud Server Component

The Cloud Server component manages the storage and processing of emergency alerts received from gateway nodes. The server records the uploaded GPS coordinates along with associated metadata such as timestamps and message identifiers. These records can then be accessed by monitoring systems or emergency responders for situational awareness and rescue coordination. By integrating with the decentralized mesh network through gateway devices, the cloud server enables large-scale monitoring while the core communication system remains operational without continuous Internet connectivity.

3.3 Module Division

3.3.1 User Interface Module

The User Interface (UI) module provides the interaction layer between the user and the emergency communication system. It allows users to control the mesh network operation and initiate communication actions during emergency situations. Through this interface, users can enable or disable the mesh service, send short text messages, and broadcast their current GPS location. The interface also displays system status indicators such as mesh network activity and Internet connectivity, along with a real-time activity log showing sent and received messages. The UI is designed to be simple, responsive, and easy to use in high-stress situations. It is implemented using Jetpack Compose, which enables a state-driven interface where updates such as incoming messages or connectivity changes

are automatically reflected on the screen.

3.3.2 Message Processing Module

The Message Processing module is responsible for preparing messages for transmission and interpreting received packets. It encodes user-generated text messages and GPS coordinates into compact data packets suitable for Bluetooth Low Energy transmission. Each packet includes a message type identifier and a Time-To-Live (TTL) value that controls how far the message can propagate through the network. When a packet is received from another device, the module decodes the message, extracts the relevant information, and determines whether the message should be displayed to the user or rebroadcasted to neighboring nodes. This module also helps prevent redundant transmissions by handling message identification and ensuring that packets are forwarded only once. By maintaining simple packet structures and lightweight processing, the module supports efficient communication on resource-constrained mobile devices.

3.3.3 BLE Communication Module

The BLE Communication module manages all device-to-device communication within the mesh network. It utilizes Bluetooth Low Energy advertising and scanning capabilities available on modern smartphones to exchange packets between nearby devices. Each node periodically broadcasts packets while simultaneously scanning for packets from neighboring devices. When a new packet is received, the module forwards it to the Message Processing module for decoding and possible rebroadcast. The communication process is maintained through a foreground service to ensure continuous operation even when the application runs in the background. The use of BLE enables low-power operation and infrastructure-independent communication, allowing smartphones to form an ad-hoc mesh network capable of supporting multi-hop message propagation.

3.3.4 Gateway Module

The Gateway module provides connectivity between the offline mesh network and external Internet-based services. In situations where a device within the mesh network obtains Internet access through Wi-Fi or mobile data, that device temporarily acts as a gateway node. The gateway detects stored or incoming GPS distress messages and forwards them

to a remote cloud server. This process follows an opportunistic store-carry-forward model inspired by delay-tolerant networking principles, where messages are carried within the mesh until a node with Internet connectivity becomes available. By allowing any device to function as a gateway when connectivity is detected, the system avoids reliance on a fixed infrastructure and increases the likelihood that emergency alerts eventually reach external systems.

3.3.5 Cloud Server Module

The Cloud Server module serves as the centralized component responsible for storing and managing emergency alerts received from gateway devices. When GPS coordinates are uploaded by a gateway node, the server records the location data along with associated metadata such as timestamps and device identifiers. These records can then be accessed by emergency responders or monitoring systems to identify the location of affected individuals. The server may also provide APIs or dashboards that allow authorities to visualize alerts on a map and coordinate rescue operations. By integrating with the decentralized mesh network through gateway nodes, the cloud server enables long-range communication and centralized monitoring while the core system remains capable of operating without Internet connectivity.

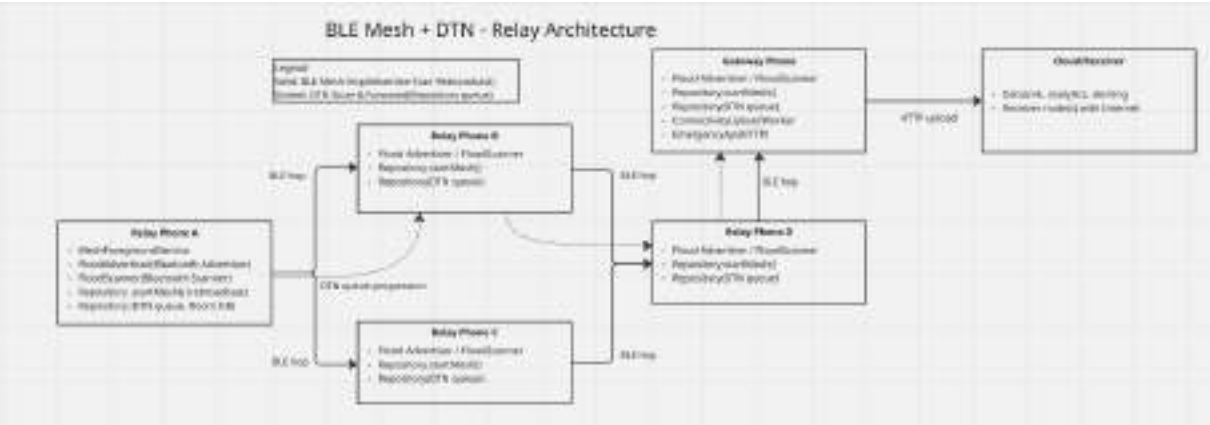


Figure 3.3: System Flow Diagram

3.4 Tools and Technologies

3.4.1 Software Requirements

Development Environment

- Android Studio: Narwhal Feature Drop 2025.1.2
- Java/JDK: 21
- Gradle (wrapper): 8.13.
- Android Gradle Plugin: 8.12.0.
- Kotlin: 2.2.0.
- Git: Git CLI + a hosted repo (GitHub/GitLab) with branch protection.

Android App – Libraries

- Nordic Bluetooth Mesh: no.nordicsemi.android:mesh:3.3.7
- Nordic BLE helpers: no.nordicsemi.android:ble:2.10.2, ble-ktx:2.10.2
- Fused Location (GPS): com.google.android.gms:play-services-location:21.3.0
- WorkManager (background "send when online"): androidx.work:work-runtime-ktx:2.10.3
- HTTP client: com.squareup.okhttp3:okhttp:5.1.0 (you set this in the catalog)
- Desugaring (Java 21 backports): com.android.tools:desugarjdklibs:2.1.5
- AndroidX core/appcompat/material: per your catalog (core-ktx 1.17.0, appcompat 1.7.1, material 1.12.0)

Cloud / Server

- Render as Endpoint: HTTPS endpoint that accepts payload.
- Auth/keys: simple API key or signed tokens (store securely).
- Storage: PostgreSQL

Web Browser Requirements

- Any modern web browser (up-to-date version)
- Supported browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari

Network Requirements

- Reliable broadband internet connection (minimum 10 Mbps recommended)

Operating System

- Windows 10/11
- macOS (latest versions)
- Linux distributions (Ubuntu 20.04 or higher recommended)

3.4.2 Hardware Requirements

Android Phones (test nodes)

- Android 8.1+ (API 27+) — matches your minSdk=27.
- BLE 4.2+ required; BLE 5.x strongly recommended (better range & stability).
- At least one phone with intermittent internet (Wi-Fi/Cell) to act as the uplink when available.

Central Processing Unit (CPU)

- Intel Core i5/i7/i9 series (8th generation or newer)
- AMD Ryzen 5/7/9 series (3000 series or newer)

Memory (RAM)

- Minimum: 8 GB RAM
- Recommended: 16 GB RAM or more for optimal performance

Storage

- Minimum: 256 GB available storage

3.5 Work breakdown and Responsibilities

3.5.1 Module 1: User Interface Module - Alan, Aaron, Aaromal

Purpose: Provides the interaction layer between the user and the emergency communication system. It allows users to start or stop the mesh network, send short text messages, broadcast GPS coordinates, and monitor system activity. The interface also displays connectivity indicators and an activity log showing both transmitted and received messages. It also includes a dashboard created to display distress signals received by the cloud server

3.5.2 Module 2: BLE Communication Module - Alan, Aaron

Purpose: Enables short-range device-to-device communication using Bluetooth Low Energy (BLE). This module manages BLE advertising and scanning processes, allowing smartphones to exchange packets and form an ad-hoc mesh network capable of multi-hop message propagation.

3.5.3 Module 3: Message Processing Module - Alan, Aaron

Purpose: Handles the encoding, decoding, and validation of packets transmitted across the mesh network. It converts user-generated text and GPS data into compact packet formats, processes incoming packets, manages Time-To-Live (TTL) values, and ensures proper message forwarding within the network.

3.5.4 Module 4: Gateway Module - Amal, Aaromal

Purpose: Acts as a bridge between the offline mesh network and the Internet. When a node detects active Internet connectivity, this module forwards GPS distress alerts received from the mesh network to a remote cloud server using secure network communication protocols.

3.5.5 Module 5: Cloud Server Module - Amal, Aaromal

Purpose: Responsible for receiving and storing emergency alerts forwarded by gateway nodes. The cloud server maintains a centralized record of GPS locations and associated metadata, enabling monitoring systems or emergency responders to access and analyze the information for rescue coordination..

3.6 Key Deliverables

- Emergency message creation and transmission.
- GPS distress message creation and transmission
- Multi-hop forwarding using BLE Mesh relays.
- Delay-tolerant, store-carry-forward message delivery under disconnections.
- Automatic synchronization to cloud servers once Internet access is restored.
- Prototype Android Application: Implements BLE Mesh communication, DTN-based message forwarding, and cloud synchronization.
- Gateway and Cloud Integration: Enables reliable message uploads and alerts via HTTPS interface.
- Cloud Hosted Dashboard: Dashboard displaying GPS distress messages received at Cloud Server
- Simulation and Field Testing Results: Evaluation under various network densities and connectivity scenarios.
- Technical Documentation: Comprehensive report including design, architecture, algorithms, and performance results.
- Demonstration Setup: Real-world demo showcasing end-to-end message delivery and system reliability.

3.7 Project Timeline

Figure 3.4: Gantt Chart

3.8 Summary of chapter

This chapter outlined the comprehensive system design of the BLE Mesh Communication System: MeshComm. It detailed the architecture, critical components, algorithms, software/hardware needs, team responsibilities and deliverables. The design emphasizes resilience, decentralization, and delay tolerance to enable reliable, infrastructure-independent emergency communication in disaster and low-connectivity scenarios.

Chapter 4

Results and Discussions

This chapter presents the results obtained from the implementation and testing of the proposed Bluetooth Low Energy (BLE) based mesh communication system. The objective of this evaluation is to analyze the functionality, reliability, and practical feasibility of the system in enabling communication without relying on conventional network infrastructure. Various tests were conducted to verify message transmission, multi-hop communication, GPS location broadcasting, and opportunistic forwarding through gateway devices with Internet connectivity. The chapter discusses the system implementation environment, the behavior of the mesh network during message dissemination, and the performance of the gateway mechanism in forwarding emergency alerts to a cloud server. The observations obtained from these experiments are analyzed to assess the strengths and limitations of the proposed approach and to demonstrate the effectiveness of the system in supporting emergency communication scenarios.

4.1 System Implementation and Testing Setup

The proposed emergency communication system was implemented as an Android application designed to operate on smartphones equipped with Bluetooth Low Energy (BLE) and GPS capabilities. The application was developed using Kotlin within the Android Studio development environment. Testing was performed on multiple Android devices supporting BLE communication to evaluate the functionality of the mesh networking system. The application was installed on each device and configured to operate in both advertising and scanning modes simultaneously.

Experiments were conducted in indoor environments to simulate typical real-world usage conditions where users may be located within buildings or shelters during emergency situations. The devices were placed at varying distances to observe the effective com-

munication range of BLE and to verify the ability of the system to propagate messages through multiple nodes. Both Internet-connected and offline scenarios were tested to evaluate the behavior of the gateway module. These experiments allowed verification of message transmission, packet forwarding, GPS broadcasting, and opportunistic gateway forwarding to the cloud server.

4.2 Message Transmission and Multi-Hop Communication Results

The implemented system successfully enabled text-based message communication between nearby devices without requiring Wi-Fi or cellular network connectivity. When a user sends a message from one device, the message is encoded into a compact packet and broadcast using BLE advertising. Neighboring devices detect the broadcast packet through scanning and forward it to other nodes within range. This process allows the message to propagate across multiple devices in a multi-hop manner.

During testing, messages were successfully transmitted between devices located beyond the direct BLE communication range by utilizing intermediate nodes. The Time-To-Live (TTL) mechanism effectively controlled the number of times a packet could be rebroadcasted, preventing excessive network flooding and reducing redundant transmissions. The system demonstrated reliable message dissemination across the test devices, confirming the feasibility of simple flooding-based routing for small-scale mesh networks. These results indicate that smartphones can form an effective ad-hoc communication network during situations where conventional infrastructure is unavailable.

4.3 GPS Alert Broadcasting and Gateway Functionality

In addition to text messaging, the system supports broadcasting GPS coordinates to communicate the location of users during emergency situations. When the GPS broadcast function is triggered, the application retrieves the device's geographic coordinates and encodes them into a packet format suitable for BLE transmission. Similar to text messages, these packets are propagated through the mesh network using the flooding mechanism.

Testing confirmed that GPS packets could be successfully transmitted across multiple devices. When a device with active Internet connectivity received the GPS packet, the gateway module detected the available network connection and forwarded the location data to the remote cloud server. This behavior demonstrates the effectiveness of the opportunistic gateway model, where messages originating from offline nodes can eventually reach external services through intermediary devices. The successful integration of mesh communication with cloud connectivity enables emergency alerts to be transmitted beyond the local mesh network.

4.4 Observation and Performance Analysis

The experimental evaluation of the system revealed several important observations regarding its performance and practical usability. The BLE flooding approach proved to be an effective and lightweight method for enabling short-range emergency communication between smartphones. The system was able to operate without relying on centralized infrastructure, making it suitable for disaster scenarios where traditional communication networks may be unavailable.

However, certain limitations were also observed. The effective communication range of BLE is relatively limited compared to Wi-Fi or cellular networks, meaning that message propagation depends on the presence of intermediate nodes within range. Environmental factors such as walls, obstacles, and interference in the 2.4 GHz spectrum may also affect communication reliability. Additionally, excessive node density could potentially lead to increased packet collisions due to the flooding-based forwarding mechanism.

Despite these limitations, the results demonstrate that the proposed system provides a practical and low-cost solution for enabling communication in emergency situations. By leveraging existing smartphone hardware and simple networking mechanisms, the system can facilitate information sharing and location reporting even when conventional communication infrastructure fails.

4.5 Conclusion

This chapter presented the implementation results and experimental evaluation of the proposed Bluetooth Low Energy (BLE) based mesh communication system. The testing process demonstrated that the system successfully enables device-to-device communication without relying on conventional network infrastructure such as cellular networks or Wi-Fi. Text messages were effectively transmitted between nearby devices, and the multi-hop forwarding mechanism allowed messages to propagate beyond the direct communication range through intermediate nodes. The Time-To-Live (TTL) based flooding approach ensured controlled packet dissemination while preventing excessive network traffic.

The results also confirmed the successful broadcasting of GPS location information through the mesh network. When a device with Internet connectivity was encountered, the gateway module was able to forward the received GPS alerts to the cloud server, validating the effectiveness of the opportunistic store-carry-forward communication model.

Chapter 5

Conclusions & Future Scope

5.1 Conclusion

The proposed system demonstrates a practical approach for enabling emergency communication in situations where conventional network infrastructure becomes unavailable. By leveraging Bluetooth Low Energy (BLE), smartphones can form a decentralized mesh network that allows devices to exchange messages and propagate information across multiple hops without relying on cellular or Internet connectivity. The implemented system supports both text message communication and GPS location broadcasting, enabling users to share distress information and coordinate assistance within the mesh network. The use of a simple flooding mechanism with a controlled Time-To-Live (TTL) ensures efficient message dissemination while preventing excessive network congestion.

Furthermore, the integration of an opportunistic gateway mechanism allows GPS alerts generated within the offline mesh network to be forwarded to a cloud server whenever a device with Internet connectivity becomes available. This capability ensures that emergency location information can reach external systems and rescue authorities even if the originating device is completely disconnected from the Internet. The successful implementation and testing of the system demonstrate the feasibility of using BLE-based mesh networking on commodity smartphones to provide a lightweight, infrastructure-independent communication platform for disaster response and rural connectivity scenarios.

5.2 Future Scope

The proposed system can be further enhanced through several extensions to improve its usability and scalability. One potential improvement is the implementation of a dedicated responder mode, which would allow rescue personnel to detect and prioritize devices

that are actively broadcasting distress GPS signals. Such a feature could help responders quickly identify and locate individuals who require assistance within the mesh network. Future work could also explore more advanced routing or forwarding strategies to reduce packet redundancy and improve efficiency in larger networks. Additional security mechanisms such as message authentication and encryption may be incorporated to protect sensitive information transmitted within the network. Finally, integration with additional wireless technologies such as Wi-Fi Direct or LoRa could extend the communication range and further enhance the reliability of the system in large-scale disaster environments.

References

[1] J. Rondón, P. Santana, and D. Carrillo, "Understanding the Performance of Bluetooth Mesh: Reliability, Delay, and Scalability Analysis," *IEEE Internet of Things Journal*, vol. 6, no. 4, pp. 7243–7255, 2019.

[2] Y. Zhao, H. Li, and K. Wang, "A Comprehensive Evaluation of Bluetooth Low Energy Mesh Networks Using Commercial Off-the-Shelf Devices," *IEEE Access*, vol. 12, pp. 134567–134580, 2024.

[3] L. Wang, J. Li, and M. Li, "A Rapid Flooding Approach Based on Adaptive Delay and Low-Power Sleep for BLE Mesh Networks," *IEEE Access*, vol. 12, pp. 65323–65332, 2024.

[4] S. Nakayima, M. Ogata, and A. Higashino, "Combining Software-Defined and Delay-Tolerant Networking Concepts With Deep Reinforcement Learning to Enhance Vehicular Networks," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 2, pp. 2556–2571, 2024.

[5] F. Longo, L. Seno, and M. Re, "Collective Victim Counting in Post-Disaster Response: A Distributed, Power-Efficient Algorithm via BLE Spontaneous Networks," *IEEE Sensors Journal*, vol. 25, no. 1, pp. 312–324, 2025.

[6] J. Li, M. Li, and L. Wang, "Constrained Flooding Based on Time Series Prediction and Lightweight GBN in BLE Mesh," *Sensors*, vol. 24, no. 14, pp. 4752–4767, 2024.

[7] A. Beben, A. Bak, and M. Sosnowski, "Efficient Relay Node Management in BLE Mesh Networks," *International Journal of Electronics and Telecommunications*, vol. 65, no. 4, pp. 633–640, 2019.

[8] S. Darroudi and C. Gomez, "Bluetooth Low Energy Mesh Networks: A Survey," *Sensors*, vol. 17, no. 7, pp. 1467–1485, 2017.

[9] S. K. Mohsin, M. A. Mohammed, and H. M. Yassien, "Developing of Bluetooth Mesh Flooding Between Source-Destination Linking of Nodes in Wireless Sensor Networks," *Eastern-European Journal of Enterprise Technologies*, vol. 6, no. 9, pp. 78–89, 2021.

[10] J. Gomez, M. Montes-de-Oca, and J. J. Camacho-Escoto, "Flood and Contain: An Optimized Repeal-Based Flooding Algorithm for Wireless Ad Hoc and Sensor Networks," *Sensors*, vol. 20, no. 20, pp. 5914–5928, 2020.

List of Publications

1 Aaromal M, Aaron Korah Mathew, Alan Madhu Kuriakose and Amal Zachariah Mathew, "RESQNET:Emergency Communication System for Disaster Response" - applied for Indian Patent

Appendix A: Presentation

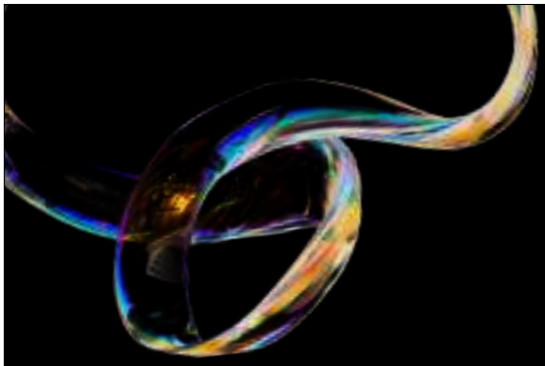
MESHCOMM: EMERGENCY COMMUNICATION SYSTEM

Group 13
Project Guide: Mr. Tonny Binoy

Presented by:
Aaromal M (U2203001)
Aaron Korah Mathew (U2203005)
Alan Madhu Kuriakose (U2203022)
Amal Zachariah Mathew (U2203035)

CONTENTS

1 Problem Definition
2 Project Objective
3 Purpose and Need
4 Literature Survey
5 Proposed Method
6 Architecture Diagram
7 Module Description
8 Results and Outputs
9 Assumptions
10 Work breakdowns and Responsibilities
11 Hardware and Software Requirements
12 Risks and Challenges
13 Conclusions
14 References



PROBLEM DEFINITION

Issues in disaster zones:

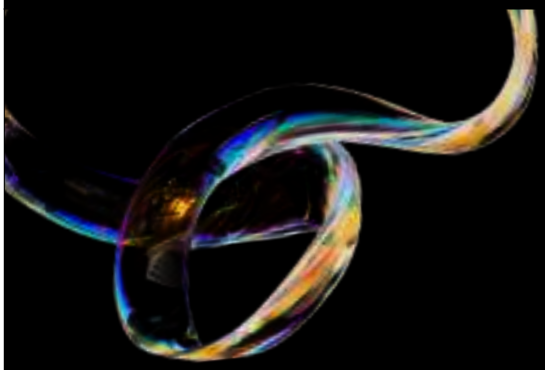
- Internet/cellular service is unavailable
- Devices are mobile and sparsely distributed
- Message Flooding causes collisions and wastes energy

Challenges

- Guaranteeing message delivery despite disconnections
- Preventing duplication and buffer overflow
- Maintaining low energy usage in battery-limited devices

Definition:

Design a decentralized, energy efficient and delay-tolerant emergency communication system using Bluetooth Low Energy with Opportunistic Gateway Finder that guarantees eventual message delivery even under sparse connectivity and network partitioning.



PROJECT OBJECTIVE

Main Goal:

- Build a resilient mobile communication system for Emergency Response that is not reliant on local Internet/cellular infrastructure

Specific Objectives

- Develop an app that allows peer-peer broadcast communication for devices that are in proximity
- Relay Emergency Communication to essential services by broadcasting GPS location supported by opportunistic gateway forwarding.
- Develop a web-based application for emergency services to track emergency messages received at a cloud server.

PURPOSE

- To develop an infrastructure-independent emergency communication system.
- To enable smartphone-to-smartphone communication using Bluetooth Low Energy(BLE).
- To provide a reliable method for sending distress messages when cellular networks fail
- To implement a lightweight flooding-based mesh network without complex routing protocols
- To support opportunistic GPS forwarding to an Internet-enabled device

NEED

- During disasters(floods, earthquakes, landslides), cellular towers and internet services often fail.
- Survivors are left without any communication channel to request help
- Existing communication systems depend heavily on centralized infrastructure.
- Laboratory-grade emergency communication systems are expensive and not widely deployable
- Smartphones are widely available and already equipped with BLE and GPS.

LITERATURE SURVEY

1. ***L. Wang, J. Li, and M. Li, "A rapid flooding approach based on adaptive delay and low-power sleep for ble mesh networks," IEEE Access, vol. 12, pp. 65323–65332, 2024.***

Control Logic: Environmental Adaptation. Decisions are made locally based on the immediate density and coverage status of neighbours.

Philosophy: Redundancy and Energy Optimization. The goal is to maximize performance while minimizing power consumption under the constraints of the managed flooding model.

Key Difference: FADLS attempts to manage when and if a node relays, preserving the simplicity of the flooding model, while ACE focuses on dictating where (which channel) the packet is sent to achieve fundamental latency gains

LITERATURE SURVEY

2. ***A. Hussein, R. Tarek, H. Osama, R. Fawzy, K. Elsayed, and M. Taha, "An aodv-based routing scheme for large-scale bluetooth low-energy mesh networks," in Proceedings of the 8th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC), Online, 2020, pp. 1–4.***

Control Logic: On-Demand Path Discovery. This logic is triggered only when a source node needs to send data and lacks a route.

Philosophy: Resource Conservation and Link Reliability. The objective is to establish a high-quality link using the fewest hops possible.

Key Difference: This is a full routing replacement, aiming to eliminate broadcast redundancy by replacing it with unicast traffic along the established path, whereas ACE focuses on optimizing the transmission channel itself

LITERATURE SURVEY

3. ***K. S. Bhandari, A. S. M. S. Hosen, and G. H. Cho, "Coar: Congestion-aware routing protocol for low power and lossy networks for iot applications," Sensors, vol. 18, no. 11, p. 3838, 2018.***

Control Logic: Multi-Criteria Decision Making (MCDM). This is a sophisticated decision process executed before a parent selection.

Philosophy: Proactive Congestion Prevention. The primary goal is network reliability during traffic surges (e.g., emergency scenarios) by maximizing node resource health.

Key Difference: CoAR focuses on optimizing the logical choice of the next-hop based on complex resource health, making it an ideal congestion supplement to any routing protocol, whereas ACE optimizes the physical transport of the packet across the spectrum.

LITERATURE SURVEY

4. *M. Wang, Y. Li, J. Lv, Y. Gao, C. Qiao, B. Liu, and W. Dong, "Ace: A routing algorithm based on autonomous channel scheduling for bluetooth mesh network," Electronics, vol. 11, no. 1, p. 113, 2022.*

ACE's methodology centres on the Autonomous Channel Scheduling (ACS) function, which operates without requiring external coordination or time synchronization. Fixed Channel Selection: Every node k calculates its single, dedicated listening channel (from 0 to 39) based on its unique hardware address.

LITERATURE SURVEY

5. *L. Davoli, M. Moreni, and G. Ferrari, "Roumble: A sink-oriented routing protocol for ble mesh networks," IEEE Internet of Things Journal, vol. 12, no. 10, pp. 14202 14218, 2025.*

Control Logic: Structural Command and Control. RouMBLE is driven by centralized, top-down control from the Sink Nodes.

Philosophy: Structural Efficiency and Data Convergence. The primary goal is to organize the field devices into a predictable structure for easy data collection.

Key Difference: RouMBLE's primary optimization is architectural (creating a fixed topology for MP2P traffic) and relies on compressed addressing to save space in the limited BLE payload, contrasting with ACE's optimization of the physical transmission medium.

LITERATURE SURVEY

6. ***A. Aijaz, A. Stanoev, D. London, and V. Marot, "Demystifying the performance of bluetooth mesh: Experimental evaluation and optimization," 2021. [Online].***

Experimental evaluations further highlighted practical challenges. Aijaz et al. conducted a comprehensive performance analysis of Bluetooth mesh networks, identifying issues like packet collisions, relay overhead, and scalability constraints [6]. Their findings emphasize the importance of bounding rebroadcast operations to balance the reliability and efficiency factors.

LITERATURE SURVEY

7. ***L. R. Paiva, A. K. Albuquerque, and G. P. Koslovski, "Combining Software-Defined and Delay-Tolerant Networking Concepts with Deep Reinforcement Learning Techniques," IEEE Latin America Transactions, vol. 17, no. 12, pp. 1954–1963, 2019.***

Beyond BLE-specific protocols, Delay-Tolerant Networking (DTN) has been extensively studied for intermittently connected environments. Paiva et al. combined the DTN concepts with software-defined networking and deep reinforcement learning to optimize message forwarding even when there is disruption [7]. While effective, such approaches introduce significant computational and architectural complexities, thereby limiting their practicality for rapid deployment on commodity smartphones during emergencies.

LITERATURE SURVEY

8. ***C. W. Khor and N. S. Ahmad, "BLE-based Indoor Localization with Temporal Convolutional Network," Journal of Engineering Research, 2025***

BLE has also been applied in location-aware systems. Khor and Ahmad demonstrated BLE-based indoor localization using temporal convolutional networks, validating the feasibility of transmitting compact location-related data over BLE [8]. Although focused on localization accuracy, their work supports the suitability of BLE for disseminating GPS-related information in constrained environments.

PROPOSED METHODOLOGY

The proposed system adopts a decentralized, infrastructure-independent communication model in which every smartphone participates equally as a sender, receiver, and relay. The methodology is based on simple BLE broadcasting combined with controlled flooding to enable multi-hop message propagation.

Message Types
Two distinct packet types are supported:

1. User-Generated Text Messages: These are packets that contain short text messages intended for local dissemination among nearby users, such as survivors or first responders.
2. GPS Coordinate Packets: These are packets that encapsulate latitude, longitude, timestamp, and a unique message identifier. They are intended to propagate through the mesh until they encounter a device with active internet connectivity.

PROPOSED METHODOLOGY

Flooding and TTL Control:
Each packet is assigned a Time-To-Live (TTL) value at creation. Upon receiving a packet, the device does the following operations:

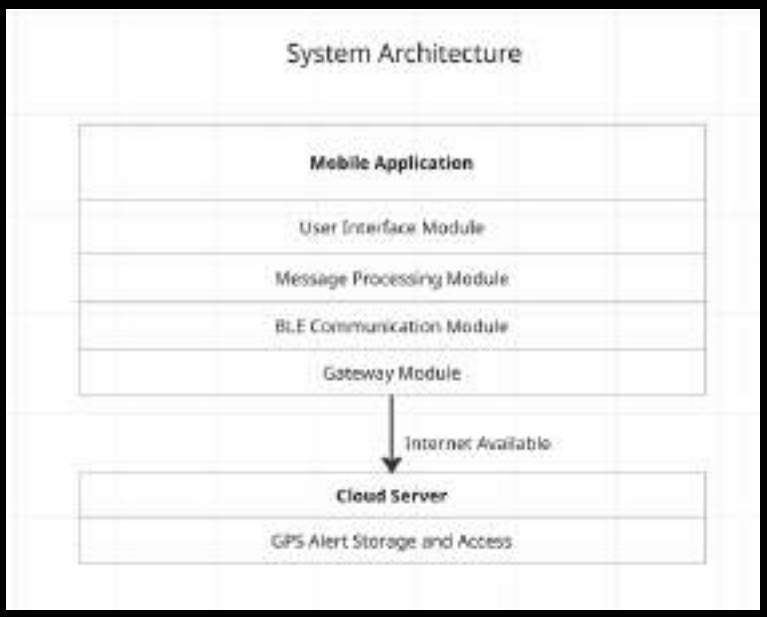
- The device checks whether the packet has been previously processed using its unique identifier.
- If the packet is new and TTL > 0, the device rebroadcasts it after decrementing the TTL value.
- If TTL value is zero, then the packet is no longer forwarded and it gets killed.
- This bounded packet flooding approach allows rapid propagation while preventing infinite rebroadcast loops and excessive channel utilization.

PROPOSED METHODOLOGY

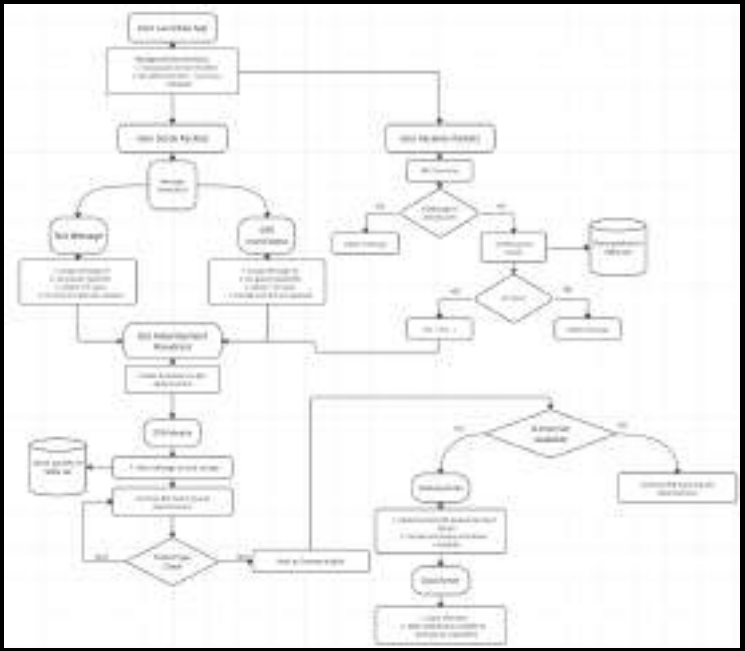
Opportunistic Gateway Forwarding:
Devices continuously monitor their network connectivity. When a device that possesses the GPS packets detect an active internet access, it transitions into a gateway mode and uploads the stored GPS alerts to a cloud server. This process requires no user intervention and enables delayed but reliable propagation of emergency information.

Continuous Operation:
To ensure an uninterrupted functionality during mobility and background operation, the system runs as an Android foreground service. BLE advertising and scanning operate concurrently, enabling simultaneous transmission and reception of packets.
This methodology emphasizes robustness, minimal state maintenance, and suitability for unpredictable disaster environments.

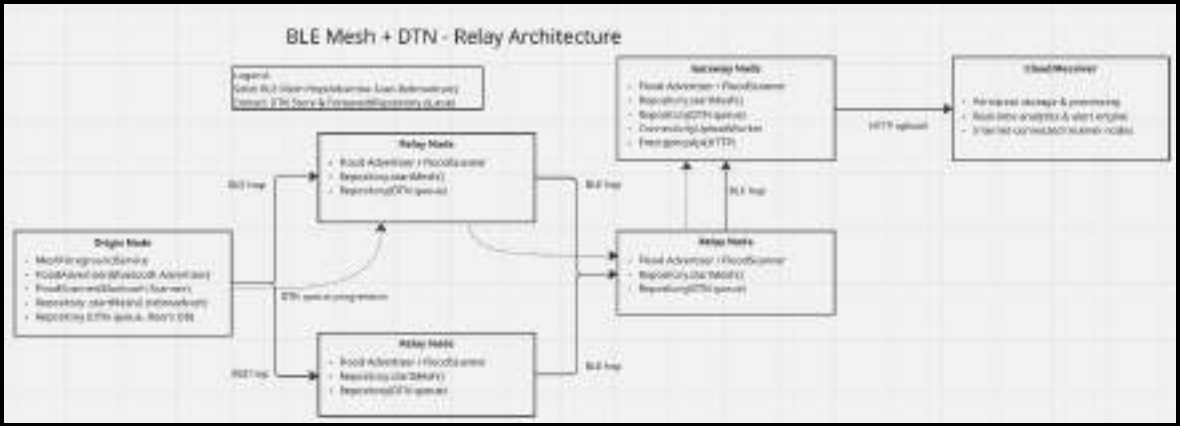
ARCHITECTURE DIAGRAM



SYSTEM FLOWCHART



SYSTEM FLOW DIAGRAM



MODULES

USER INTERFACE MODULE

Purpose: Provides an interactive interface for users to send and receive emergency messages

Key Responsibilities:
- Mesh ON/OFF toggle control
- Text message input and transmission
- GPS broadcast trigger button
- Display of sent and received messages
- Internet connectivity status indicator
- Real-time activity log

Features:
- Built using Jetpack Compose
- State-driven UI updates
- Clean, minimal emergency-focused design
- Works in offline mode

Importance: Ensures ease of use of high-stress emergency situations

MODULES

MESSAGE PROCESSING MODULE

Purpose: Handles encoding, decoding and classification of messages.

Key Responsibilities:
- Packet formatting (Text / GPS types)
- TTL (Time-To-Live) handling
- Message deduplication
- Flooding logic control
- Packet validation and parsing

Components
- TextCodec (text encoding/decoding)
- GpsCodec (coordinate encoding/decoding)
- PacketType classification
- TTL decrement before rebroadcast

Importance: Ensures reliable multi-hop propagation without routing tables

MODULES

BLE COMMUNICATION MODULE

Purpose: Enables device-to-device communication using Bluetooth Low Energy

Key Responsibilities:
- BLE Advertising (broadcast packets)
- BLE Scanning (receive packets)
- Foreground service for continuous operation
- Multi-hop flooding
- Low-power operation

Characteristics:
- Infrastructure-independent
- Works without Wi-Fi or mobile data
- Supports dynamic topology changes
- Resilient to node mobility

Importance: Forms the core of the network layer of the system

MODULES

GATEWAY MODULE

Purpose: Acts as a bridge between offline network and the Internet

Key Responsibilities:
- Detect Internet connectivity (Wi-Fi / Mobile Data)
- Receive GPS packets from mesh
- Store-carry-forward mechanism
- Upload distress alerts to cloud server
- Prevent duplicate uploads

Operational Model:
- Opportunistic forwarding
- Delay-Tolerant Networking (DTN) inspired
- No fixed routing required

Importance: Ensures that signals eventually reach rescue authorities

MODULES

CLOUD SERVER MODULE

Purpose: Stores and manages emregency alerts received from gateway devices

Key Responsibilities:

- Receive uploaded GPS alerts
- Store timestamp and location
- Provide dashboard/API access
- Enable rescue coordination
- Maintain alert history

Features:

- Real-time monitoring dashboard
- Map-based visualization
- Alert prioritization
- Notification system

Importance: Enables large-scale coordination and centralized monitoring

App UI

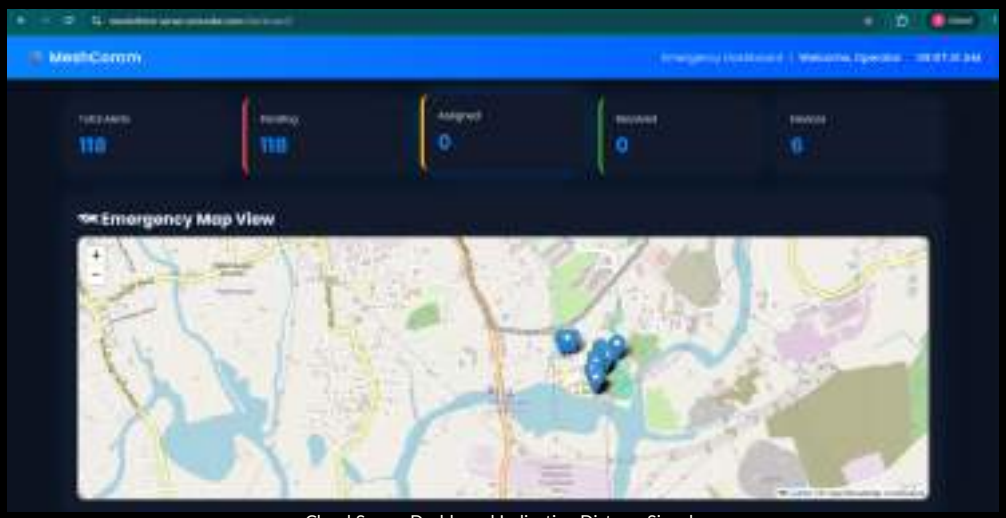
RESULTS AND OUTPUT

The prototype implementation was evaluated using multiple Android smartphones under conditions with no Wi-Fi or mobile data connectivity. Devices were carried by users moving independently, emulating realistic disaster or evacuation scenarios.

- Multi-hop propagation was successfully achieved, with messages relayed across multiple devices using BLE broadcasts.
- TTL control effectively limited unnecessary rebroadcasts while maintaining sufficient propagation range.
- GPS packets were reliably delivered to the cloud once a gateway device obtained internet access, even when the originating device never regained connectivity.
- Energy consumption remained low due to BLE usage and short broadcast payloads.
- The system remained operational under frequent device movement and intermittent contact patterns.

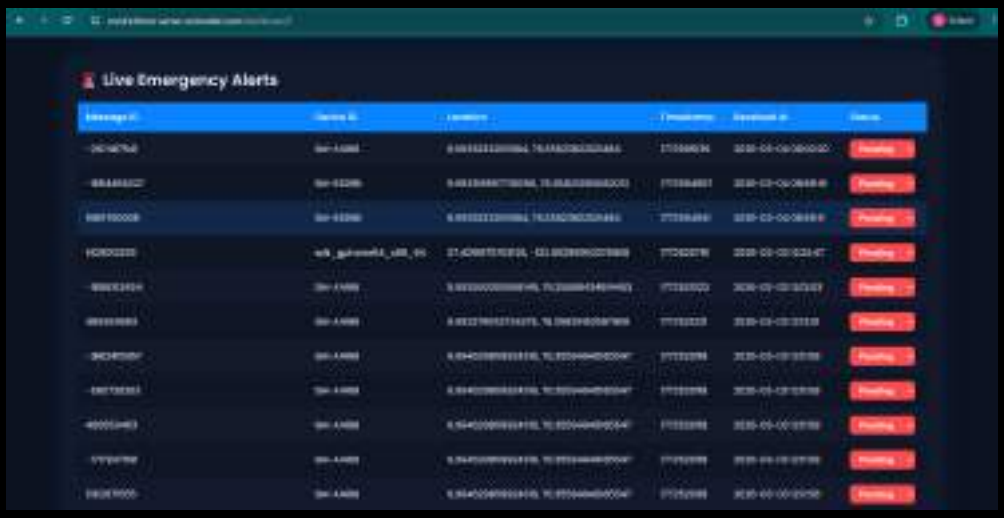
These observations confirm that simple BLE flooding can provide reliable offline communication without requiring complex routing logic or infrastructure.

RESULTS AND OUTPUT



Cloud Server Dashboard Indicating Distress Signals

RESULTS AND OUTPUT



Cloud Server Dashboard Indicating Distress Signals

ASSUMPTIONS

Device Assumptions

- All participating devices are smartphones equipped with:
  - BLE
  - GPS
- The application is installed and running on all nodes.
- Users grant required permissions(bluetooth,location,notification)

Operational Assumptions

- Users manually activate the mesh service during emergencies
- Message sizes are small (text and GPS coordinates)
- Devices have sufficient battery power during operation
- Flooding with bounded TTL is sufficient for small to medium-scale scenarios

Network Assumptions

- Atleast one device in the network may eventually gain Internet access.
- Devices are withing BLE communication range(5-30 meters)
- Nodes are mobile but periodically come in to contact(store and carry forward model)

Environmental Assumptions

- Disaster scenarios involve partial or complete failure of cellular infrastructure
- The network operates in low-resourse and infrastructure-less environments.
- The system is intended for emergency and short-term deployment

WORK BREAKDOWN AND RESPONSIBILITIES

AAROMAL M

- Cloud Server Module
- Gateway Module

AARON KORAH MATHEW

- User Interface Module
- Message Processing Module

ALAN MADHU KURIAKOSE

- BLE Communication Module
- Message Processing Module

AMAL ZACHARIAH MATHEW

- Gateway Module
- Cloud Server Module

HARDWARE REQUIREMENTS

- Android smartphone with BLE and GPS
- Internet-enables devices to act as gateway node
- Cloud server for alert storage

SOFTWARE REQUIREMENTS

- Android Studio + Kotlin
- Jetpack Compose
- BLE APIs
- Google Location Services
- Cloud REST API

RISKS AND CHALLENGES

Technical RIsks

- Limited BLE Range
- Message collisions in dense networks
- Energy drain due to frequent scanning

Operational Risks

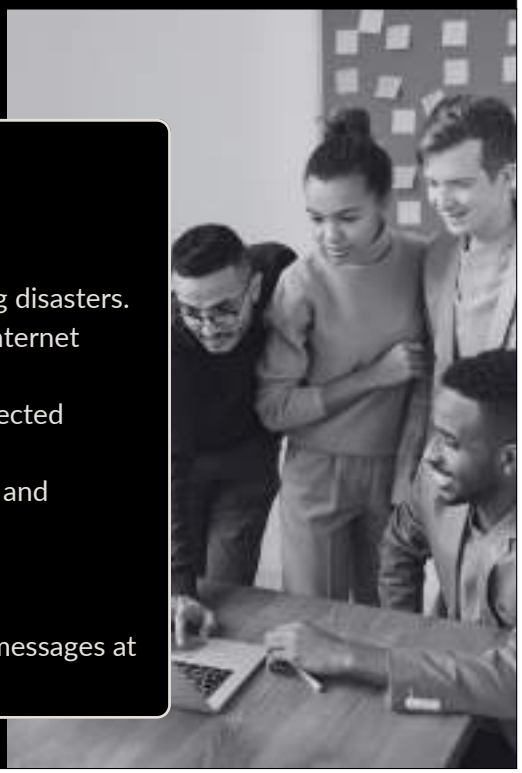
- Users may not keep the app running in background.
- Risk of false alerts from malicious nodes.

Research Challenges

- Balancing reliability vs energy efficiency.
- Designing scalable DTN buffers for large networks.

CONCLUSION

- Proposed a resilient communication system during disasters.
- Uses BLE (multi-hop) + DTN (delay-tolerance) + Internet Gateway forwarding.
- Ensures eventual delivery even in sparse, disconnected networks.
- Applicable to disaster relief, rural communication, and defense scenarios.
- Future scope:
  - IoT sensor alerts
  - AI-based priority measurement for incoming messages at server.



REFERENCES

1. L. Wang, J. Li, and M. Li, “A rapid flooding approach based on adaptive delay and low-power sleep for ble mesh networks,” IEEE Access, vol. 12, pp. 65323–65332, 2024.
2. A. Hussein, R. Tarek, H. Osama, R. Fawzy, K. Elsayed, and M. Taha, “An aodv-based routing scheme for large-scale bluetooth low-energy mesh networks,” in Proceedings of the 8th International Japan-Africa Conference on Electronics, Communications, and Computations (JAC-ECC), Online, 2020, pp. 1–4.
3. K. S. Bhandari, A. S. M. S. Hosen, and G. H. Cho, “Coar: Congestion-aware routing protocol for low power and lossy networks for iot applications,” Sensors, vol. 18, no. 11, p. 3838, 2018.
4. M. Wang, Y. Li, J. Lv, Y. Gao, C. Qiao, B. Liu, and W. Dong, “Ace: A routing algorithm based on autonomous channel scheduling for bluetooth mesh network,” Electronics, vol. 11, no. 1, p. 113, 2022.
5. L. Davoli, M. Moreni, and G. Ferrari, “Roumble: A sink-oriented routing protocol for ble mesh networks,” IEEE Internet of Things Journal, vol. 12, no. 10, pp. 14202 14218, 2025.
6. A. Aijaz, A. Stanoev, D. London, and V. Marot, “Demystifying the performance of bluetooth mesh: Experimental evaluation and optimization,” 2021. [Online].
7. L. R. Paiva, A. K. Albuquerque, and G. P. Koslovski, “Combining Software-Defined and Delay-Tolerant Networking Concepts with Deep Reinforcement Learning Techniques,” IEEE Latin America Transactions, vol. 17, no. 12, pp. 1954–1963, 2019.
8. C. W. Khor and N. S. Ahmad, “BLE-based Indoor Localization with Temporal Convolutional Network,” Journal of Engineering Research, 2025.
9. Darroudi, S. & Gomez, C. "Bluetooth Low Energy Mesh Networks: A Survey". Sensors, MDPI, 2020.
10. Beben, A. et al. "Enhancing Reliability and Stability of BLE Mesh Networks". Sensors, MDPI, 2024.

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of

mathematics, natural sciences and engineering sciences

3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. Individual and Team work: Function effectively as an individual, and as a member or leader in teams, and in multidisciplinary settings.

10. Communication: Communicate effectively with the engineering community and with society at large. Be able to comprehend and write effective reports documentation.

Make effective presentations, and give and receive clear instructions.

11. Project management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team. Manage projects in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

Course Outcomes

S.No	Description	Bloom's Taxonomy Level
1	Apply engineering fundamentals and domain knowledge to identify and model real-world problems relevant to the project.	Apply
2	Conduct literature survey and analyze existing research work to understand current solutions and identify research gaps.	Apply
3	Demonstrate teamwork and effective communication while performing requirement analysis and system design.	Apply
4	Design and implement appropriate methodologies and tools to develop solutions for the identified problem.	Apply
5	Analyze experimental results, interpret findings, and prepare technical documentation for the developed system.	Analyze
6	Present technical content effectively through written reports and oral presentations while demonstrating ethical and professional responsibility.	Apply

CO-PO AND CO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C01	2	2	2	1		2	2	1	1	1	1	2	2	2	2
C02	2	2	2		1	3	3	1	1	1	1	2	2		
C03								3	2	2	1				3
C04			2		3	2	2	3	2			3	3		
C05	2	3	3	1	2						1	3			
C06				2		2	2	3	1	1		3			

Justifications for CO-PO-PSO Mappings

Mapping	Justification
CO1-PO1	Applies engineering fundamentals to model and solve real-world problems.
CO1-PO2	Involves problem analysis and identification of solutions.
CO1-PO3	Requires system and design development to meet requirements.
CO1-PO4	Includes limited experimentation and investigation during system design.
CO1-PO5	Uses modern tools, simulation software and development platforms.
CO1-PO6	Solutions may impact societal and industrial contexts.
CO1-PO7	Considers sustainability during solution development.

Mapping	Justification
CO1-PO8	Ethical considerations are taken during system implementation.
CO1-PO9	Team collaboration is required during modelling and development.
CO1-PO10	Communication of modelling and development results.
CO1-PO11	Resource planning and project scheduling are required.
CO1-PO12	Encourages independent learning and domain knowledge integration.
CO1-PSO1	Ability to identify, analyse and design solutions for technically and economically feasible problems.
CO1-PSO2	Applying algorithms and software engineering practices for development.
CO1-PSO3	Computer science fundamentals can be applied in competitive research problems.
CO2-PO1	Apply mathematics, science and engineering fundamentals while identifying relevant literature.
CO2-PO2	Identify, formulate and analyse complex engineering problems through literature review.
CO2-PO3	Design solutions based on insights obtained from the literature survey.
CO2-PO5	Use modern tools while exploring and analysing related solutions.
CO2-PO6	Apply appropriate techniques and resources to analyse the literature.
CO2-PO8	Apply ethical principles while reviewing and using research work.
CO2-PO9	Literature survey activities require teamwork and coordination.
CO2-PO10	Communicate survey results effectively with the engineering community.
CO2-PO11	Demonstrate understanding of engineering management principles during literature study.
CO2-PO12	Encourages independent and life-long learning through research.
CO2-PSO2	Application of domain-specific tools and technologies for development.
CO3-PO9	Function effectively as an individual and as a member or leader in teams.

Mapping	Justification
CO3-PO10	Communicate effectively with engineering community and society.
CO3-PO11	Demonstrate engineering management understanding during requirement analysis.
CO3-PO12	Ability to engage in independent learning and adapt to technological change.
CO3-PSO3	Students apply computer science fundamentals while preparing technical reports.
CO5-PO2	Analyse complex engineering problems while preparing technical documentation.
CO5-PO3	Develop and design solutions for complex problems.
CO5-PO4	Understand design of experiments and interpretation of results.
CO5-PO5	Use appropriate tools while preparing technical documentation.
CO5-PO6	Connect engineering solutions with societal needs.
CO5-PO9	Develop communication skills during report preparation.
CO5-PO11	Demonstrate project management and planning skills.
CO5-PO12	Encourage life-long learning through research and documentation.
CO5-PSO1	Apply computer science fundamentals for research and development.
CO5-PSO2	Ability to use domain tools and techniques during technical work.
CO6-PO1	Study fundamentals of computer science while organizing technical content.
CO6-PO3	Analyse complex engineering problems and present findings clearly.
CO6-PO4	Interpret data and experimental results effectively.
CO6-PO5	Use modern tools for preparing reports and presentations.
CO6-PO8	Apply ethical principles while communicating scientific findings.
CO6-PO9	Develop teamwork and collaborative communication skills.
CO6-PO10	Demonstrate effective written and oral communication skills.
CO6-PO12	Recognise importance of life-long learning in technical communication.

Mapping	**Justification**
CO6-PSO1	Identify and communicate multidisciplinary engineering solutions.
CO6-PSO2	Apply programming and development skills while documenting work.
CO6-PSO3	Apply computer science fundamentals in research and communicate findings effectively.